

Set A

29.05.2024
I

JECRC UNIVERSITY
End Term Examination- 2023-24

B.Tech. -II Semester

Subject: (DMA002A) Engineering Mathematics-II (Common to all)

Time: 3 hours

Maximum Marks: 100

Instructions: (1) Attempt all the questions

(2) Mention the appropriate Question number before your answers

Course Outcomes: At the end of the course, the student should be able to:

CO1: Use matrices, determinants and techniques for solving systems of linear equations in the different areas of Linear Algebra. Understand the definitions of Vector Space and its linear Independence. Solve Eigen value problems and apply Cayley Hamilton Theorem.

CO2: Understanding convergence of sequence and series.

CO3: Identify, analyze and subsequently solve physical situations whose behavior can be described by First order and first degree ordinary differential equations and higher order linear differential equation with constant coefficients.

CO4: Determine solutions to second order linear differential equations with variable coefficients.

CO5: Understanding the series solutions of second order linear differential equations with variable coefficients.

Part -A

5*2=10

A1[CO1] Find the rank of the following matrix, $A = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 3 \\ 1 & 3 & 4 & 1 \end{bmatrix}$.

A2[CO2] Define monotonic increasing and monotonic decreasing sequence.

A3[CO3] Solve: $(D^3 - 9D^2 + 26D - 24)y = 0$.

A4[CO4] Check the given differential equation $x^2 \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + y = \frac{1}{(1-x)^2}$ is exact or not.

A5[CO5] Solve the PDE by Lagrange's method, $yzp + zxq = xy$.

Part-B

5*7=35

B1[CO1] Test the consistency, for the systems of equations and if possible then solve,
 $x + y + z = 8$, $2x + 3y + 2z = 19$, $4x + 2y + 3z = 23$.

B2[CO2] Test the convergence of the series $\sum_{n=1}^{\infty} \frac{(2n-1)}{n(n+1)(n+2)}$.

B3[CO3] Solve: $\frac{dy}{dx} - 4y = 2x - 4x^2$.

B4[CO4] Solve $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = x^2 \log x$.

B5[CO5] Solve the PDE by Charpits method, $px + qy = pq$.

Part-C

5*11=55

C1[CO1] Find the Eigen values and Eigen vectors of the following matrix- $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$.

C2[CO2] Test the convergence of the series, $\sum_{n=0}^{\infty} \frac{(2n+1)}{n^3+1} x^n$.

C3[CO3] Solve: $(D^2 + 9)y = e^x \sin^2 x$.

C4[CO4] Solve: $\frac{d^2 y}{dx^2} + 4y = 4 \tan 2x$ using method of variation of parameter.

C5[CO5] Solve in series: $(2 - x^2) \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 2y = 0$.